



EV Charging Station Performance Analysis

Using Microsoft Excel, Power Query and Oracle SQL

DATA ANALYST PORTFOLIO PROJECT

Mohan Prasanth M · Data Analyst

Project Introduction

Electric vehicle charging stations continuously generate operational information — including station availability, charging utilization, waiting time, pricing, weather conditions, traffic conditions and charger status. This project analyzes large-scale EV charging operational data to generate actionable business insights using Microsoft Excel, Power Query and Oracle SQL.

Dataset Source Public EV Charging Dataset	Total Records 1,317,750
Charging Stations 150	Cities 15
Charging Networks 8	Average Utilization 41.62%
Average Waiting Time 2.57 Minutes	Average Charging Price 0.31

The dataset represents operational charging station information collected across multiple cities and charging providers. The project demonstrates an end-to-end analytics workflow beginning from raw CSV data through data validation, data cleaning, SQL analysis and business reporting.

Technology Stack

Five tools were used across the end-to-end analytics workflow — from raw CSV data through data validation, cleaning, SQL analysis and business reporting. Each technology served a specific purpose in handling the 1.3 million-record dataset.



Microsoft Excel

Used for data inspection and workbook preparation. Excel served as the initial environment for reviewing the raw CSV dataset before loading into Power Query and the Excel Data Model.



Power Query

Used to connect the CSV file, validate column types and load data into the Excel Data Model. Power Query was necessary because the dataset exceeded Excel worksheet limitations.



Oracle Database

Used to store all 1.3 million records. Oracle Database provided a scalable and reliable environment for managing the large-scale EV charging operational dataset.



Oracle SQL Developer

Used for importing data and executing SQL scripts. Oracle SQL Developer served as the primary interface for running validation, cleaning and analysis queries against the database.

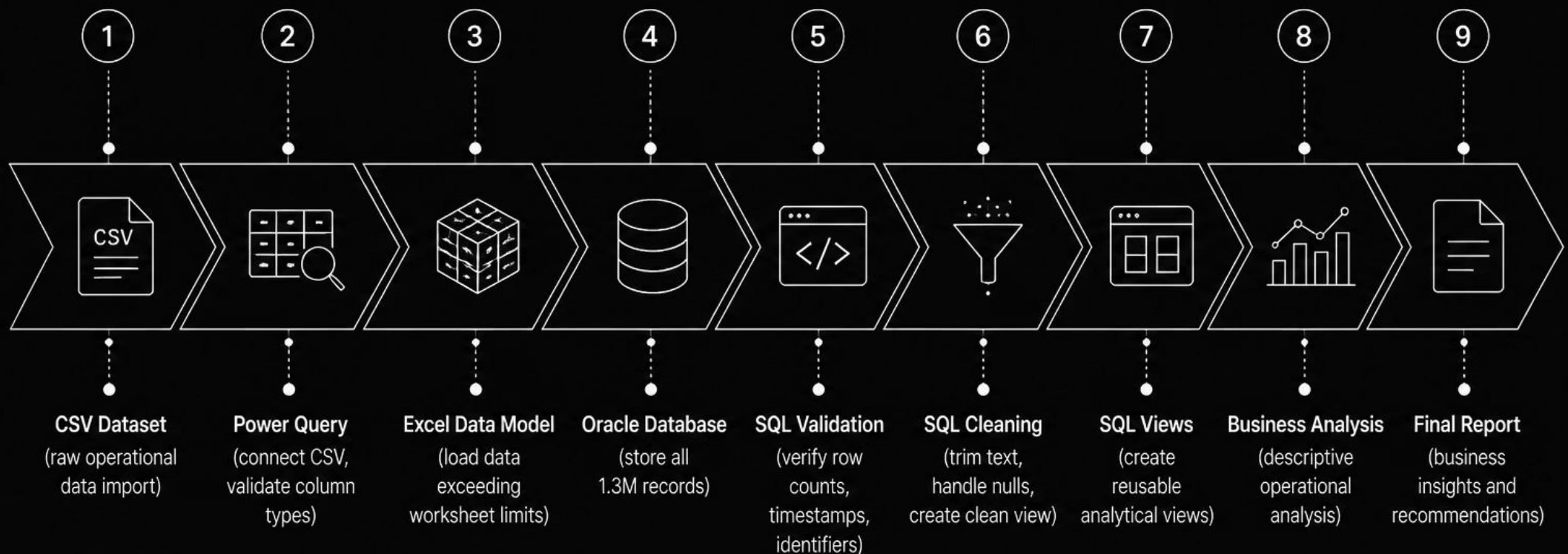


Oracle SQL

Used for validation, cleaning, grouping, aggregation, analytical views and business analysis. SQL queries transformed raw operational data into structured, analysis-ready datasets.

Project Workflow

The project demonstrates a complete end-to-end analytics workflow — beginning from raw CSV data through data validation, data cleaning, SQL analysis and business reporting. Each stage transforms the data progressively, ensuring quality and accuracy before generating final business insights.



The workflow begins with the raw CSV dataset, which is connected and validated through Power Query before being loaded into the Excel Data Model. Data is then imported into Oracle Database, where SQL validation confirms row counts, timestamps, station identifiers, city values, charging network values and numeric columns — verifying successful import without losing records. SQL cleaning trims text values, handles null values, validates timestamps and creates a reusable clean view, producing an analysis-ready dataset. SQL views simplify reporting and business analysis. The final stage delivers business insights and recommendations based on the operational analysis.

Data Validation & Cleaning

Data Validation

The following validation activities were performed after importing the dataset into Oracle Database to ensure data integrity and accuracy across all 1.3 million records:

- Validated row count after import — confirmed all 1,317,750 records were successfully imported without losing any records
- Validated timestamp format across all session records
- Validated station identifiers for consistency and completeness
- Validated city values across all 15 cities in the dataset
- Validated charging network values across all 8 networks
- Validated numeric columns including utilization, waiting time and pricing fields

Data Cleaning

After validation, the following cleaning activities were performed to prepare an analysis-ready dataset:

- Trimmed text values to remove leading and trailing whitespace
- Handled null values appropriately across all fields
- Validated timestamps for format consistency and accuracy
- Created a reusable SQL clean view (**VW_EV_SESSIONS_CLEAN**) to encapsulate all cleaning logic
- Prepared a clean, analysis-ready dataset for downstream SQL analysis and business reporting

❏ The primary table **FACT_EV_SESSIONS** stores all 1,317,750 session records. The reusable clean view ensures consistent data quality across all subsequent analysis.

SQL Architecture & Concepts

A structured SQL architecture was implemented using a primary fact table and five reusable analytical views. These views simplify reporting and business analysis by pre-aggregating data at the session, city, network and station levels.

Primary Table

FACT_EV_SESSIONS

Primary fact table storing all 1,317,750 EV charging session records – the foundation for all downstream analysis.

SQL Views

VW_EV_SESSIONS_CLEAN

Cleaned and validated session data

VW_EV_SUMMARY

Overall project summary metrics

VW_EV_CITY_SUMMARY

City-level performance aggregation

VW_EV_NETWORK_SUMMARY

Charging network performance

VW_EV_STATION_SUMMARY

Individual station analysis

SQL Concepts Applied

The following SQL concepts were applied to perform descriptive business analysis across the EV charging dataset:

- **Data Retrieval:** SELECT, WHERE, ORDER BY
- **Aggregation:** GROUP BY, HAVING, COUNT, SUM, AVG, MIN, MAX
- **Conditional Logic:** CASE statements
- **Date Handling:** Date Functions for timestamp analysis
- **Database Objects:** Views for reusable analytical layers, Indexes for query performance optimization

These SQL concepts enabled comprehensive descriptive business analysis – from overall project summaries to detailed charging station, network, city, utilization, pricing, waiting time, peak hour, weekend, weather and traffic analysis. Reusable SQL views were created to simplify reporting and ensure consistent results across all analytical queries.

Business Analysis

Reusable SQL views were created to simplify reporting and enable comprehensive business analysis across all dimensions of EV charging operations. The following ten analytical areas were explored using Oracle SQL aggregate functions, grouping and analytical views:



Overall Project Summary

High-level metrics across all 1.3 million session records, 150 stations, 15 cities and 8 charging networks.



Charging Station Analysis

Performance metrics aggregated at the individual station level using VW_EV_STATION_SUMMARY.



Charging Network Analysis

Comparative performance across all 8 charging networks using VW_EV_NETWORK_SUMMARY.



City Performance Analysis

Operational metrics broken down by city using VW_EV_CITY_SUMMARY across 15 cities.



Charging Utilization Analysis

Analysis of the 41.62% average utilization rate across stations and networks.



Pricing Analysis

Examination of the average charging price of 0.31 across networks and cities.



Waiting Time Analysis

Analysis of the 2.57-minute average waiting time across stations and peak periods.



Peak Hour Analysis

Identification of high-demand charging periods using SQL date functions.



Weekend Analysis

Comparison of charging patterns and utilization between weekdays and weekends.



Weather & Traffic Analysis

Operational impact assessment based on weather conditions and traffic data.

Verified Project Results

The project successfully processed and analyzed a large-scale public EV charging dataset. All results were verified through SQL validation queries confirming data integrity across the full 1,317,750-record dataset.

1.3M

Total Records

1,317,750 EV charging session records successfully imported and validated without data loss.

150

Charging Stations

Operational data analyzed across 150 charging stations.

15

Cities

Multi-city analysis covering 15 cities and 8 charging networks.

8

Networks

Comparative network performance analysis across 8 charging providers.

41.62%

Avg. Utilization

Average charging station utilization rate across the full dataset.

2.57

Avg. Wait (Min)

Average waiting time in minutes across all charging sessions.

0.31

Avg. Price

Average charging price per session across all networks and cities.

📄 All metrics were derived through Oracle SQL aggregate queries (COUNT, SUM, AVG, MIN, MAX) executed against the validated FACT_EV_SESSIONS table and reusable analytical views.

Project Deliverables & Business Insights

Project Deliverables

The following deliverables were produced as part of this end-to-end analytics project:

- **Excel Workbook** — Data inspection, Power Query transformations and workbook preparation
- **Oracle Database** — Full storage of all 1,317,750 records with optimized schema
- **Oracle SQL Scripts** — Complete validation, cleaning and analysis query scripts
- **SQL Views** — Five reusable analytical views for streamlined reporting
- **Business Report** — Comprehensive operational analysis and findings
- **Project Documentation** — Full methodology and workflow documentation

Business Insights

Key analytical outcomes achieved through this project:

- Successfully imported and validated more than 1.3 million records without data loss
- Created reusable SQL analytical views to simplify ongoing reporting
- Generated city, network and station-level summaries for operational decision-making
- Performed comprehensive operational analysis using Oracle SQL
- Improved understanding of charging utilization (41.62%) and waiting time (2.57 min) patterns
- Demonstrated efficient handling of large datasets using Excel Power Query and Oracle Database

Business Recommendations

- Increase charging capacity in highly utilized locations
- Monitor waiting time regularly to maintain service quality
- Improve charger availability during peak demand periods
- Schedule maintenance during low-demand periods
- Monitor charging network performance continuously
- Continue periodic operational analysis for ongoing improvement

Conclusion

This project demonstrates practical, production-ready skills in Microsoft Excel, Power Query and Oracle SQL using a real-world dataset containing more than 1.3 million EV charging session records. The end-to-end analytics workflow — from raw CSV data through validation, cleaning, SQL analysis and business reporting — reflects the type of work expected in entry-level Data Analyst, Business Analyst and SQL Analyst roles.

Skills Demonstrated

- Microsoft Excel & Power Query
- Oracle SQL & SQL Developer
- Data Validation & Cleaning
- SQL Query Writing & Aggregate Functions
- Views & Indexes
- Business Analysis & Reporting
- Problem Solving & Analytical Thinking

Role Suitability

This project is suitable for **entry-level Data Analyst, Business Analyst and SQL Analyst roles** because it demonstrates:

- Data validation and quality assurance at scale
- SQL querying, aggregation and analytical view creation
- Business reporting with actionable insights and recommendations
- Analytical thinking applied to a real-world operational dataset
- Efficient handling of large datasets using industry-standard tools

Mohan Prasanth M

Data Analyst

SQL | Microsoft Excel | Oracle SQL

Thank you for reviewing this portfolio project.